

# Logan MacFarland

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Data Scientist with 9+ years of experience developing and deploying production machine learning systems across various domains, including bioinformatics, e-commerce, and finance. I am excited to apply the skills I've gained across my data science experience back to my roots of bioinformatics. Most of my career has been allowing me to be the data specialist for small teams, translating unanswered business questions to deployment ready data based solutions. Experienced across the full ML lifecycle from data gathering, feature engineering and cloud deployment. Currently completing a Master's in Data Science (Aug. 2026).

## Experience

### *Work Study @ Dreher Laboratory, Oct 2013 - Jun 2015*

- Engineered cloud-based pipelines for automated processing of terabyte-scale genomic datasets, improving workflow scalability and reproducibility.
- Led development of computational workflows for the lab's Oxford Nanopore MinION sequencing platform, supporting long-read sequencing and downstream analysis.
- Designed and evaluated sequence alignment methods using novel algorithms and sequencing chemistries to improve genomic analysis quality.
- Developed methods combining PacBio long-read and Illumina short-read sequencing to resolve alignment gaps and generate more complete, accurate genome assemblies.

### *Data Scientist @ Goldstar Events, Mar 2019 - Feb 2022*

- Developed a production recommendation system using an advanced factorization machine architecture that incorporated regional trends and user behavior while requiring minimal historical user data. Models increased global click-through rate by approximately 20%.
- Built an item similarity model using gradient boosting, NLP features, and behavioral data to improve content discovery and user engagement.
- Designed end-to-end machine learning infrastructure including feature engineering, data pipelines, memory optimizations, and AWS deployment.
- Led data integration efforts following acquisition by developing text-processing algorithms to identify and merge duplicate event listings.

### *Founding Engineer / CTO @ [Riise.ai](https://riise.ai), Aug 2024 - Jan 2025*

- Led development of an end-to-end AI platform for estimating commercial building retrofit value using computer vision and public datasets.

- Built and deployed PyTorch computer vision models for commercial building analysis using satellite imagery and public datasets.
- Designed Django backend APIs supporting ML inference, report generation, and geospatial data processing.
- Built React frontend with Google Maps visualization.
- Integrated LLM-powered chatbot capable of report generation and natural-language querying our backend to work as a user interface to our data.
- Managed architecture across frontend, backend, cloud infrastructure, and ML systems.

#### Data Scientist @ First Bank of the Lake, *Mar 2023 - July 2023*

- Developed an XGBoost model to estimate loan lifespan, enabling more informed purchasing decisions in the secondary loan market.
- Engineered features from internal and public datasets to generate robust predictions despite incomplete or inconsistent source data.
- Integrated SQL databases, XML files, and third-party web data into automated model pipelines deployed through AWS SageMaker.
- Collaborated with business stakeholders to optimize loan bidding strategies using historical market performance and model recommendations.

#### Data Scientist @ Credibility Capital, *July 2022 - Feb 2023*

- Designed and implemented a centralized AWS data lake using Lambda, AppFlow, and Glue to support machine learning workflows across multiple data sources.
- Built automated data pipelines between AWS and Google Cloud, improving data accessibility and reducing manual processing.
- Optimized cloud infrastructure and resource utilization to reduce operational costs by approximately 30% while improving system reliability.
- Designed SQL databases, indexing strategies, and automation scripts to support scalable analytics and model development.

#### Data Analyst @ Nike, *Jan 2019 - Mar 2019*

- Developed a Python and R fuzzy matching algorithm to reconcile addresses across large enterprise datasets with inconsistent formatting.
- Built scalable data pipelines and confidence scoring methods using hand-labeled training data to improve matching accuracy.
- Automated address reconciliation workflows, significantly reducing manual review while improving data quality.

## Education

Willamette University - *Master's in Data Science (expected 08/2026)*

Oregon State University - *Bachelor's in Microbiology*